1. Epidemiology and Impact of Hypoglycemia on Patients with Diabetes

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Epidemiology

The significance of hypoglycemia as an adverse effect of treatment became apparent within a few years of the introduction of insulin (1). As the importance of maintaining glucose concentrations as close to normal as possible began to accumulate, the emerging relationship between tight glucose control with subcutaneous insulin therapy and the risk of severe hypoglycemia caused some to question the safety of the approach (2). These doubts were silenced by the publication of the Diabetes Control and Complications Trial (DCCT), but the epidemiological data that accompanied the DCCT emphasized the risk from hypoglycemia in those undertaking intensive insulin therapy (3). The data also highlighted the need for a good understanding of the adverse effects of therapy when agreeing upon clinical approaches with patients or their relatives.

Systematic epidemiological data began to emerge in the 1980s as hypoglycemic adverse effects became more apparent. The evidence was initially obtained from overnight monitoring of individuals with type 1 diabetes (4) and retrospective questioning of those attending hospital clinics, and it was assumed at the time that the risks of hypoglycemia were generally confined to this group.

These data suggested that therapy (usually with twice daily insulin) in patients with type 1 diabetes led to severe hypoglycemia in around 10% to 30% of patients per year (5). The risk of severe hypoglycemia was also reported as a rate, usually the number of overall number of episodes experienced by each patient per year, although both methods failed to highlight that rates of hypoglycemia are not normally distributed. Most individuals with either type 1 or insulin-treated type 2 diabetes experience few or no episodes at all, while a small proportion experience hypoglycemia, even severe episodes, very frequently (6). This type of distribution requires fairly sophisticated statistical techniques to identify factors that may predict the likelihood of further episodes

By whatever way hypoglycemic burden is measured, the field has been hampered by a range of different definitions. This has made comparison between different studies challenging, and the resulting heterogeneity has limited the use of meta-analysis in combining data from different studies to estimate the benefit of new insulins and technologies such as insulin pumps (7). The American Diabetes Association convened a working group in an attempt to overcome these difficulties (8). Their work endorsed a definition of severe hypoglycemia as an episode that produces cognitive impairment, sufficient to require the help of another person to recover. However, the definition of mild or biochemical hypoglycemia continues to be debated (9, 10).

Using the definition of severe hypoglycemia as an episode requiring the help of another person, a large clinical questionnaire from Denmark completed by 411 patients with type 1 diabetes reported a frequency of mild symptomatic episodes of 1.8 episodes per patient per week compared to a frequency of severe episodes of 0.027 episodes per patient per week (11). The latter equated to around 1.6 severe hypoglycemic attacks per year, and during their lives around a third of patients had been comatose on at least 1 occasion. Remarkably, this rate had not changed when examined in a similar survey 10 years later, despite the introduction of more modern regimens together with the use of insulin analogues (12). In the DCCT, of those randomized to standard therapy (1 to 2 injections per day with little monitoring), around 10% experienced at least 1 episode over 12 months (13). The risk was 3 times greater in the group undertaking intensive therapy where the glucose targets were close to normal. In the DCCT, rates of hypoglycemia rose exponentially as HbA1c approached near normal levels (14).

Clinical trials have generally reported rates of hypoglycemia in subjects with insulin-treated type 2 diabetes that are lower than in trials of those with type 1 diabetes with comparable levels of glycemic control. In the United Kingdom Prospective Diabetes Study (UKPDS) (15) over the first 10 years, the proportion of patients suffering a major hypoglycemic event per year was 0.4% for chlorpropamide, 0.6% for glibenclamide and 2.3% for insulin at HbA1c levels comparable to the DCCT.

The overall mean values hide the important observation that rates of hypoglycemia rose as the duration of insulin therapy increased. Furthermore, in other observational studies involving individuals with type 2 diabetes, if patients are matched for duration of insulin treatment, then rates are comparable (16). This suggests that increasing duration of type 2 diabetes is associated with an increased risk of severe hypoglycemia, perhaps due to a progressive failure of endogenous insulin secretion.

This hypothesis is supported by the results of a recent study in which rates of self-reported and biochemical hypoglycemia were compared over 9 to 12 months in individuals with differing durations of diabetes and with similar levels of glucose control (HbA1c around 7.5%)(Figure 1-1). Those with type 2 diabetes within 2 years of starting insulin had rates of hypoglycemia (both biochemical and symptomatic, including severe) that were similar to those patients who were taking sulfonylureas (17). However, patients with type 2 diabetes treated with insulin for over 5 years (and whose endogenous insulin production as measured by stimulated C-peptide secretion was lower), had significantly higher rates of hypoglycemia including severe episodes. This group had a risk of hypoglycemia that was comparable to individuals with recently diagnosed type 1 diabetes. Those with a diabetes duration of over 15 years were at greatest risk, with around 50% experiencing a severe episode over the duration of the study. Other population-based studies have also indicated that in insulin-treated individuals with type 2 diabetes, severe hypoglycemic episodes occur at rates comparable to those

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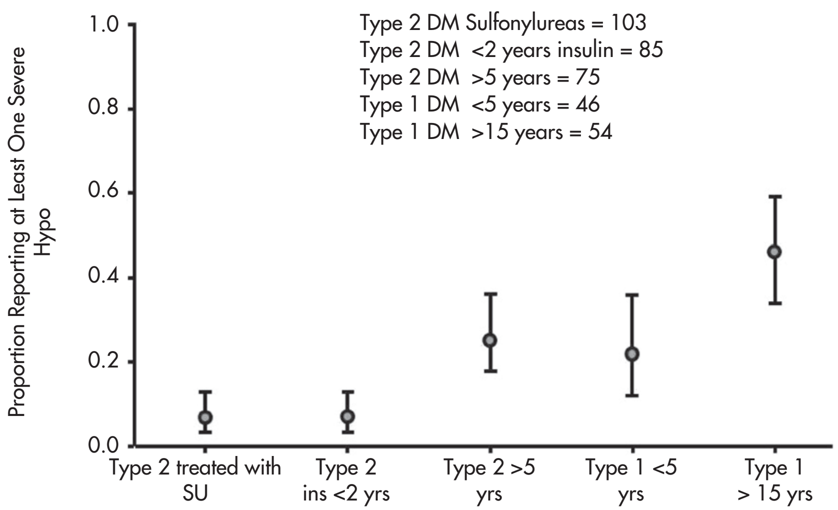

FIG 1-1. Proportion of each group experiencing at least one severe self-reported hypoglycaemic episode during 9-12 months of follow-up. Vertical bars, 95% confidence intervals (from UK Hypoglycaemia Study Group: Risk of hypoglycaemia in types 1 and 2 diabetes: effects of treatment modalities and their duration. Diabetologia 50:1140-1147, 2007, figure 2).

in type 1 diabetes (18). Thus, as far more people with type 2 diabetes are treated with insulin, the clinical problem is correspondingly greater.

Predictors of Hypoglycemic Risk

Predictors of Hypoglycemia

Episodes of hypoglycemia do not have a normal distribution. Although the factors that contribute to a high risk of hypoglycemia are common to many individuals with diabetes, some people experience very few episodes, while others are disabled by frequent severe events without any warning. While epidemiological studies have identified some of the predictive factors, much remains poorly understood, particularly the contribution of psychological and behavioral factors. Since these determine the ability of individuals to self-manage their condition, they are probably crucial.

Long Duration of Diabetes

Duration of diabetes is a powerful determinant of hypoglycemic risk. Those with a duration of diabetes of over 15 years have a 3 times higher risk of experiencing a severe episode compared to those recently diagnosed (17). The dependence on exogenous insulin and largely absent glucagon response to hypoglycemia in those lacking the ability to secrete endogenous insulin are clearly relevant factors. This is probably also why individuals with insulin-treated type 2 diabetes experience a persistent increase in risk as the duration of insulin treatment increases (19).

Impaired Awareness of Hypoglycemia

Impaired awareness of hypoglycemia is common; around 1 in 5 individuals with type 1 diabetes has difficulty recognizing hypoglycemia (20), and among patients with insulin-treated type 2 diabetes, around 1 in 10 complains of some degree of unawareness (21). Unsurprisingly, loss of awareness cosegregates with defective counter-regulatory responses, particularly the hypoglycemic activation of the sympathoadrenal system (22). Those with impaired awareness are at considerable risk of severe hypoglycemia, with a relative increase in risk of between 5 and 7 times those who can recognize hypoglycemia during standard treatment (23, 24).

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Renal Impairment

Renal impairment is an important although under-reported contributor to hypoglycemia. It impairs clearance of insulin and the metabolites of sulfonylureas. The renal contribution to gluconeogenesis (up to 30% under some conditions) and glucose reabsorption of filtered glucose are also relevant. The risk of severe hypoglycemia may be raised by as much as 2 to 3 times in individuals with type 2 diabetes who have an estimated glomerular filtration rate (eGFR) less than 60 mL/min (25). Measurement of renal function is particularly important among patients with type 2 diabetes who experience a severe episode.

Increasing Age and Cognitive Impairment

Individuals who lack the ability to recognize and treat hypoglycemia are unsurprisingly at greater risk of severe episodes. This may explain why the elderly, particularly those with cognitive impairment, are particularly vulnerable. However, the association of cognitive impairment with hypoglycemia (26) may also reflect a reduced ability to treat episodes appropriately.

Increasing age also leads to impaired sympathoadrenal responses during hypoglycemia (27), with alterations in the thresholds for both cognitive impairment and onset of symptoms reducing the margin of safety for self-treatment (28).

Psychosocial Factors

The role of psychosocial factors in determining the risk of hypoglycemia is increasingly recognized (29). Inappropriate responses of patients to raised glucose concentrations and their overcorrection with additional doses of insulin may contribute to loss of hypoglycemia awareness and the risk of subsequent severe episodes. The potential of anticipating and identifying these attitudes and developing interventions to address them is currently the increasing focus of experimental work.

Physiological Defenses to Hypoglycemia

Because individuals with insulin-treated diabetes (and those taking sulfonylureas) are unable to suppress insulin secretion as glucose levels fall below 4 mmol/L (72 mg/dL), they lack the major physiological defense that prevents hypoglycemia and must rely on other processes to mitigate the glucose-lowering effects of insulin. However, although these additional defenses are particularly important to individuals with diabetes, they are already impaired within just a few months of diagnosis in individuals with type 1 diabetes and also decline (albeit more slowly) with increasing duration in those with type 2 diabetes.

Inhibition or prevention of glucagon release impairs glucose recovery by around 40%, but if both glucagon and epinephrine are blocked, then glucose recovery is totally inhibited.

Counter-regulation

Hypoglycemia stimulates increases in glucagon and epinephrine, leading to hepatic glucose release, initially from glycogenolysis, with an increasing proportion due to gluconeogenesis. In addition, peripheral glucose uptake is progressively inhibited, partly directly, but also secondary to increases in circulating nonesterified fatty acids following epinephrine-stimulated lipolysis. Inhibition or prevention of glucagon release impairs glucose recovery by around 40%, but if both glucagon and epinephrine are blocked, then glucose recovery, at least in the experimental setting, is totally inhibited. Other hormones including growth hormone and cortisol also raise blood glucose over time, and during severe hypoglycemia, the liver releases glucose independent of hormonal control (hepatic autoregulation) (30). The virtual failure of glucose recovery from experimental hypoglycemia when both glucagon and sympathoadrenal responses are blocked suggests that other mechanisms do not play a major physiological role during acute episodes.

Symptoms of Hypoglycemia

Data establishing how patients recognize hypoglycemia have emerged both from studies in the laboratory where hypoglycemia has been induced (22, 31) and from questioning larger numbers of patients and using the statistical technique of factor analysis (32)(Table 1-1).

Adults with type 1 diabetes appear to experience 2 specific groups of symptoms, autonomic and neuroglycopenic, while other symptoms such as malaise, nausea, and headache are classified as nonspecific. Importantly, experimental studies have generally corroborated these clinical observations. Activation of the autonomic nervous system as glucose falls below normal warns patients of impending hypoglycemia, producing symptoms of sweating, palpitations, and tremor (hence the term autonomic). Other symptoms reflect increasing cerebral dysfunction as neuronal glucose supply falls below a critical level (neuroglycopenia), and include confusion

TABLE 1-1. Classification of symptoms of hypoglycemia using Principal Components Analysis in patients with insulin-treated diabetes depending on age group

Children (pre pubertal)	Adults	Elderly
Autonomic/neuroglycopenic	Autonomic	Autonomic
	Neuroglycopenic	Neuroglycopenic
Behavioral	Non-specific malaise	Neurological

and loss of concentration. They are caused by cerebral dysfunction due to reduced glucose availability (neuroglycopenia). The combination of neuroglycopenic and autonomic symptoms, on which individuals learn to rely, varies between and within the same individual, but the symptoms are crucial in alerting patients. Most patients rely on a combination to help them recognize and treat an episode before cognitive impairment progresses to a stage where they need the help of another person. Factor analysis indicates that while children can perceive autonomic and neuroglycopenic symptoms, they also experience symptoms related to alterations in emotion and behavior (33).

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While it is to be expected that parents learn to recognize hypoglycemia in their children before the children themselves, it is noteworthy that partners and other family members often alert adults to impending hypoglycemia (34). Factor analysis applied to data obtained from elderly patients with type 2 diabetes showed a greater prominence of less-specific neurological symptoms (incoordination, ataxia, and visual abnormalities) (35), suggesting an increased susceptibility of other areas of the brain to hypoglycemia in this age group.

How do Patients Recognize a Falling Blood Glucose?

In nondiabetic individuals, activation of the autonomic nervous system, manifested by an increase in plasma epinephrine (and preceded by release of glucagon) occurs at around 65 mg/dL (3.6 mmol/L) of glucose. Symptoms such as tremor or sweating are generated at around 60 mg/dL (3.3 mmol/L) and this is reflected in laboratory settings at a similar glucose level by subjectively rated symptom scores. Experimental studies have generally confirmed clinical observations that cerebral cognitive function (using sensitive tests such as 4 choice reaction time) begins to deteriorate at around 55 mg/dL (3 mmol/L) as blood glucose falls and cerebral glucose delivery diminishes (22) (Figure 1-2). Awareness of hypoglycemia depends upon patients recognizing their own peripheral autonomic response and taking action before their cognitive dysfunction becomes established in the absence of cerebral glucose delivery. Since both symptom generation (through activation of the autonomic nervous system) and cerebral dysfunction reflect neuroglycopenia, it is unsurprising that both responses occur at comparable low glucose levels. It emphasizes how important it is for patients to respond to warning symptoms and implies that a shift in activation of the sympathoadrenal response to a lower blood glucose could prevent recognition of hypoglycemia.

Physiological defenses to hypoglycemia are intact at diagnosis but become progressively impaired as diabetes duration increases. Other factors may also influence the rate of decline of these defenses, and there appears to be considerable variation in the extent and severity of impaired responses.

Impaired Endocrine Defenses and Impaired Hypoglycemia Awareness

The physiological defenses described previously are intact at diagnosis (36) but become progressively impaired as diabetes duration increases. Other factors may also influence the rate of decline of these defenses, and

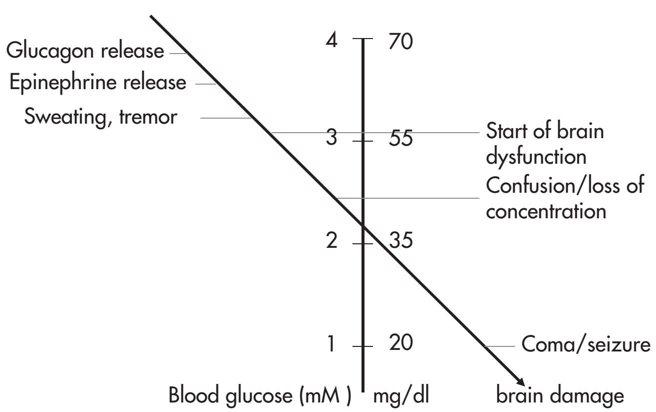

FIG 1-2. Glucose thresholds for the normal physiological response to hypoglycemia.

there appears to be considerable variation in the extent and severity of impaired responses. Moreover, those with widespread defects develop a marked decline in hypoglycemia awareness and a subsequently increased risk of severe episodes.

Glucagon

The glucagon response to hypoglycemia appears to be intact at diagnosis but begins to fail within a few months (37). Nearly all patients with type 1 diabetes exhibit absent or reduced responses after 5 years of disease. The defect is afferent; the α-cell fails to recognize hypoglycemia as a stimulus for secretion, since glucagon is secreted normally in response to other secretagogues such as arginine. There is increasing evidence in people that failing glucagon responses to hypoglycemia represent disruption of a paracrine mechanism of local insulin inhibition by adjacent α-cells as a result of their autoimmune destruction (38, 39).

Epinephrine and Sympathoadrenal Responses

Impaired epinephrine responses during hypoglycemia occur commonly among individuals with type 1 diabetes but generally develop after a longer duration of diabetes compared to impaired glucagon release. They reflect a more general defective response of sympathoadrenal activation during hypoglycemia, so that in addition to diminished circulating epinephrine levels, symptoms of sweating (sympathetic cholinergic), palpitations, and tremor are also reduced. The prevalence of impaired epinephrine or of sympathoadrenal responses to hypoglycemia is more variable than that of glucagon, although many individuals with type 1 diabetes demonstrate defective responses. A survey of published research estimated that a diminished epinephrine response was present in around 40% of patients with type 1 diabetes, most of whom had had diabetes for over 15 years (40). Impaired responses appear to be the result of a resetting of the glycemic threshold for activation to a lower blood glucose level.

Patients who are unable to release glucagon and epinephrine during hypoglycemia have a high risk of severe episodes during subsequent treatment (41). This reflects not only impaired counter-regulatory mechanisms, but probably a failure of peripheral sympathoadrenal responses, which alert patients to hypoglycemia by generation of autonomic symptoms.

Defects in the autonomic response to hypoglycemia may not only be confined to impaired secretion. There is evidence of altered β-adrenergic sensitivity in some patients with type 1 diabetes although to what extent these changes contribute to impaired hypoglycemia awareness and susceptibility to severe hypoglycemia in the clinical situation is yet to be established (42, 43).

Impaired Hypoglycemia Awareness

It has long been realized that a proportion of patients with insulin-treated diabetes have difficulty identifying impending hypoglycemia (44), but estimates of its incidence and prevalence are varied. This is a consequence of differing definitions and because awareness of hypoglycemia is rarely an all-or-nothing phenomenon. Many patients have experienced a single or occasional severe hypoglycemic episode without warning, but most of the time they are alerted by their symptoms and take appropriate action. Yet clinical surveys of the topic have identified impaired hypoglycemia awareness as a relatively common problem among individuals with type 1 diabetes, affecting as many as a quarter of adults attending hospital clinics, a proportion which reaches nearly 50% in those with a duration of diabetes over 20 years (11, 12, 45). Since impaired hypoglycemia awareness reflects diminished autonomic responses, it is unsurprising that those affected are vulnerable to severe hypoglycemia (Figure 1-3); risks of severe episodes have been reported in 2 prospective studies as varying between 3 and 7 times greater compared to those who claimed to recognize hypoglycemia (23, 24).

Strict Glycemic Control, Antecedent Hypoglycemia and Hypoglycemia-Associated Autonomic Failure

Studies in the 1980s showed that strict glycemic control and intensive insulin therapy involving individuals with type 1 diabetes resulted in reduced endocrine and symptomatic responses to hypoglycemia (46), which were associated with a resetting of the glucose threshold at which they were activated (47). This was followed by research that established that just a few hours of mild experimental hypoglycemia reduces sympathoadrenal, endocrine, and symptomatic responses to further episodes of hypoglycemia. The effect has been demonstrated both in nondiabetic individuals (48, 49) and in adults with both type 1 (50) and type 2 diabetes (51). Once induced, defective responses appear to be maintained for at least a week in nondiabetic subjects (52), although the maximum duration of effect demonstrated in individuals with type 1 diabetes is only 2 days (53).

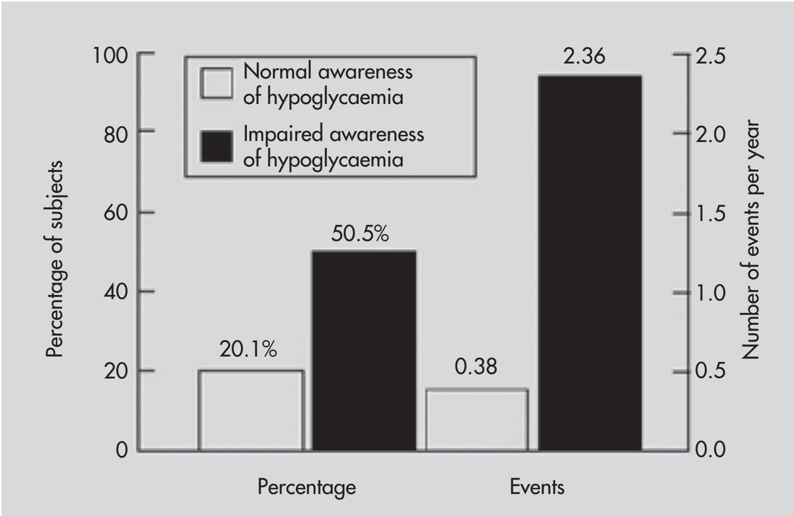

FIG 1-3. Prevalence and incidence of severe hypoglycemia (SH) in the year preceding a survey of 518 adults with Type 1 diabetes with and without impaired awareness of hypoglycemia (reproduced from Geddes et al, 2008) (20).

The extent to which antecedent hypoglycemia contributes to the widespread impairment in physiological defenses to hypoglycemia demonstrated by patients is still not entirely clear. However, it may largely account for the failure of counter-regulatory mechanisms and loss of hypoglycemia awareness after tightening glycemic control, presumably alongside those associated with long duration of disease.

The term hypoglycemia-associated autonomic failure (HAAF) has been proposed by Cryer to describe the phenomenon, and he has highlighted how an initial period of hypoglycemia could lead to a downward vicious spiral of progressively impaired physiological responses, increasing clusters of hypoglycemic episodes and the development of impaired hypoglycemia awareness (54). However, the extent to which autonomic dysfunction (a classical complication of diabetes generated by a different mechanism) may contribute is uncertain. Interestingly, antecedent exercise results in impaired sympathoadrenal responses to experimental hypoglycemia, and antecedent hypoglycemia impairs subsequent sympathoadrenal activation during exercise (55). Furthermore, antecedent hypoglycemia has been demonstrated to impair some cardiac autonomic reflexes, including baroceptor sensitivity and sympathoadrenal responses, to lower negative body pressure in nondiabetic individuals (56). On the other hand, some patients with clear evidence of classical autonomic neuropathy apparently exhibit normal counter-regulatory responses (57, 58). Some have speculated that the development of impaired defenses represents an adaptation to stress, which in the context of therapeutic hypoglycemia, increases the vulnerability of patients to subsequent episodes.

The Mechanism Underlying Impaired Hypoglycemia Awareness

Impaired hypoglycemia awareness occurs due to a change in the relationship between the physiological responses, which generate symptoms and the deterioration in cognitive ability, preventing them from being recognized. The precise mechanism has not been established, but one hypothesis utilizes the clinical and experimental observations, described previously, that patients with impaired hypoglycemia awareness exhibit diminished autonomic responses due to a resetting of the threshold for the onset of autonomic symptoms at a lower glucose concentration (59). Proponents believe that the weight of evidence indicates that the glycemic threshold for the onset of cognitive impairment is relatively fixed (60, 61). They suggest that by the time patients with diminished awareness start to generate peripheral autonomic changes such as sweating and tremor, they have already become cognitively impaired. This prevents them from responding appropriately, and unless they receive assistance, their blood glucose will continue to fall, leading to increasing confusion, coma, or seizure. Not all studies have indicated that the threshold for cognitive dysfunction is fixed; some have reported alteration of glucose thresholds to a lower level in a similar fashion to autonomic defenses (62, 63). Nevertheless, even this work indicates that the glycemic thresholds for significant cognitive impairment, when measured by sensitive tests of cognitive function, are set at higher glucose levels than those for sympathoadrenal activation.

Are Defects in Endocrine Counter-Regulation and Impaired Hypoglycemia Awareness Reversible?

The observations that a few hours of antecedent hypoglycemia lead to major alterations in physiological responses to hypoglycemia suggests that these defects may be functional rather than structural and so amenable to reversal. Clinical studies have confirmed that impaired hypoglycemia awareness, present in some cases for some years, can be reversed, at least in part with restoration of symptoms of hypoglycemia, and in some reports, resetting of glycemic thresholds at higher levels for epinephrine release and for symptoms. In some studies, investigators demonstrated that the effect of hypoglycemia avoidance was to reset the threshold for release of epinephrine to a glucose concentration above that for cognitive impairment (64, 65). Not all studies have demonstrated restoration of epinephrine responses (66), although, interestingly, symptom scores during hypoglycemia improved despite the reversal program having failed to restore epinephrine release.

The potential to reverse some aspects of diminished hypoglycemia awareness by avoiding hypoglycemic episodes offers a practical solution to the problem of loss of awareness. The work described above suggests that if patients can completely eliminate hypoglycemia for just a few weeks, they will obtain useful clinical benefit. The published studies do not provide detailed descriptions of the clinical techniques that were used to avoid hypoglycemic episodes in order to restore normal awareness. Nevertheless, the relevant features appear to include close and continuing communication with specialist medical and nursing staff and discussing blood glucose targets based on intensive monitoring. Patients need to be prepared to accept occasional high glucose values and not to overcorrect with extra doses of insulin. It is important to emphasize that the approach is not designed to run high blood glucose values but to avoid hypoglycemic episodes, and at least in the published studies, it was possible to achieve this without a major deterioration in glycemic control.

Clinical Consequences

Morbidity of Hypoglycemia

Neurological Consequences

Around 25% of all episodes of severe hypoglycemia result in coma (67), which is the principal neurological consequence of severe hypoglycemia. Coma and hypoglycemia-induced seizures can result in serious morbidity such as fracture-dislocations, soft tissue injuries, and head injury. When electroencephalography (EEG) is performed during hypoglycemia in resting subjects who are awake, typical changes that occur include a decrease in alpha waves, an increase in theta waves, and increased bursts of delta waves over the cerebral cortex, which are more evident over the anterior part of the brain (68). The EEG abnormalities are more pronounced in people with type 1 diabetes than in nondiabetic subjects, and epileptiform activity is more common. The theta wave changes persist after normoglycemia has been restored. Hypoglycemia-induced EEG changes may persist for several days and can become permanent, particularly after recurrent severe hypoglycemia. This may confound interpretation of the EEG when attempting to exclude idiopathic epilepsy. Epileptiform seizures can also provoke cardiac arrhythmias (69, 70), which might cause sudden death in association with a hypoglycemia-induced seizure. Because hypoglycemia often causes dizziness, lightheadedness, confusion, and mental obtundation, accidents causing personal injury are a common sequela (71), and may result in falls with fractures or joint dislocations, particularly in frail elderly people who may have other comorbidities such as osteoporosis and coronary heart disease.

Permanent focal neurological lesions following acute hypoglycemia are rare, whereas transient and reversible neurological deficits are relatively common; in individual cases, short-lived structural changes have been demonstrated using sophisticated neuroimaging (72, 73). Functional changes may therefore occur within the brain without leaving a permanent cerebral abnormality. Based on anecdotal clinical observation, it is thought that several hours of exposure to profound neuroglycopenia are required to produce permanent brain damage. Such a catastrophic outcome is rare and is usually the consequence of attempted suicide with deliberate massive insulin overdose (74) or associated with excessive alcohol consumption (75). Structural abnormalities observed in the brains of survivors of profound hypoglycemia include cortical and hippocampal atrophy and ventricular dilatation. These severely brain-damaged individuals have evidence of widespread cognitive impairment or exist in a vegetative state. When hypoglycemic coma is prolonged, predicting prognosis can be difficult. Elevation of serum markers of brain damage, neurone-specific enolase and S-100, within 24 to 48 hours of the onset of coma, usually indicates a fatal outcome (76).

Total cerebral blood flow is increased during acute hypoglycemia, and blood flow is altered acutely within different regions of the brain; it increases in the frontal cortex, presumably as a compensatory mechanism to preserve the supply of glucose to this part of the brain, which is very sensitive to glucose deprivation. These regional vascular changes become permanent in people who are exposed to recurrent severe hypoglycemia and in those with impaired awareness of hypoglycemia; they are then observed during normoglycemia (77). This probably represents an adaptive response of the brain to recurrent exposure to neuroglycopenia. However, the permanent hypoglycemia-induced changes in regional cerebral blood flow may promote localized neuronal ischemia, particularly if the cerebral circulation is already compromised by the development of cerebrovascular disease. This risk of cerebral ischemia may then be augmented if the individual is exposed to some other form of hemodynamic stress that affects the cerebrovascular circulation. Transient ischemic attacks and hemiplegia are recognized manifestations of hypoglycemia, and in the elderly they may be misdiagnosed as cerebrovascular disease.

Cognitive function, which includes several forms of mental activity, rapidly becomes impaired when blood glucose declines below 3.0 mmol/L (54 mg/dL). Complex and attention-demanding cognitive tasks, and those that require speeded responses are much more affected by hypoglycemia than simple tasks or those that do not have any time restraint (78, 79). Although the overall speed of response of the brain to make decisions is slowed, accuracy is preserved for many tasks (80, 81). Although many aspects of mental performance become impaired during hypoglycemia, individual differences exist in the levels at which impairment commences and in the magnitude of dysfunction that occurs. Recovery of cognitive function does not occur immediately after the blood glucose returns to normal, but in some cognitive domains may be delayed for up to 60 minutes (79), which has practical consequences for the performance of tasks that require complex cognitive functions, such as driving. The profound and negative effects that hypoglycemia exerts on mood and emotion are often unrecognized by clinicians (82). Fear of hypoglycemia is common and is a major obstacle to maintaining strict glycemic control both in type 1 and insulin-treated type 2 diabetes (83).

In adults, the long-term effect of repeated exposure to severe hypoglycemia on cognitive function is less clear. The longitudinal follow-up of the DCCT showed no differences in cognitive function between the intensive and standard treatment arms over a period of up to 20 years, despite a greater frequency of severe hypoglycemia in the intensive arm (84). Similarly, data from a systematic meta-analysis did not show any relationship between recurrent hypoglycemia and performance in cognitive tests (85). However, the brain of the very young or elderly person with diabetes may be much more susceptible to hypoglycemia-induced brain injury (86). Exposure to hypoglycemia in children with early onset of type 1 diabetes has been associated with lower verbal IQ (87), and early exposure to severe hypoglycemia in young children with type 1 diabetes appears to have adverse long-term effects on cognitive function (88). Cognitive development may be affected adversely by exposure to hypoglycemia, particularly when accompanied by induced seizures, in very young children (89). It is possible that repeated exposure to hypoglycemia may accelerate the onset of dementia in elderly people with type 2 diabetes, but as yet this remains unproven.

Mortality

Linking Mortality to Hypoglycemia

It is challenging to calculate accurately the risk of mortality due to hypoglycemia in diabetes. Death certificates are notoriously unreliable in attributing cause of death, with coding often left to hospital clerks who rarely have appropriate training or access to clinical details (4).

Sustained carbohydrate metabolism continuing postmortem, manifested by glycogenolysis, raises the blood glucose concentration in major right-sided vessels. Thus, normal or high glucose values on the right side of the heart do not exclude antemortem hypoglycemia. In contrast, glucose is also metabolized peripherally; thus low blood glucose is often found after death in those without diabetes.

The difficulties are reflected in a wide range of published estimates on hypoglycemic mortality rates. Reports from Scandinavia have estimated hypoglycemic mortality of between 7% and 10% among young adults, only slightly lower than mortality rates attributed to diabetic ketoacidosis (90, 91). Because hypoglycemia is common, however, an individual episode is unlikely to cause death.

Dead in Bed Syndrome

Since glucose is an essential neuronal fuel, as glucose falls below a critical value, reduced glucose delivery leads to cerebral dysfunction and cognitive impairment and could go on to cause permanent cerebral damage that has the potential to be fatal. As described previously, death due to hypoglycemic cerebral damage has been described but is rare. Cases are identified by prolonged coma and characteristic histological findings (92).

There is increasing evidence that hypoglycemia might contribute to mortality through its effects on the cardiovascular system. A series of 22 unexpected deaths across the United Kingdom among young patients with type 1 diabetes highlighted a strong link to nocturnal hypoglycemia (93). Victims were discovered in an undisturbed bed having gone to bed apparently well, which led to the term, dead in bed, to describe the syndrome in an accompanying editorial (94). Patients had a variable duration of diabetes, but were under 40 years of age, generally aiming for tight glycemic control and subject to nocturnal hypoglycemia.

The initial report has since been followed by other observational studies, including more recently, detailed autopsies (95) and genetic analyses (96). Another recent series involving 2 large registries in the Pittsburgh area of the United States concluded that this mode of death was increased tenfold compared with nondiabetic individuals and associated with male sex, a high HbA1c and insulin dose, and low body mass index (97).

In the nondiabetic population, the most frequent cause of sudden death is coronary heart disease and associated fatal cardiac arrhythmia. However, in a young age group where coronary disease is rare (even in individuals with diabetes), then additional factors must contribute. Since there is a link between dead in bed syndrome and hypoglycemia, it is relevant that there are plausible potential mechanisms by which hypoglycemia might contribute to mortality. Case reports of arrhythmias provoked by hypoglycemia include atrial arrhythmias and premature ventricular contractions as well as ventricular tachycardia (98–100). Furthermore, hypoglycemia causes abnormal cardiac repolarization, which itself increases the risk of sudden death. Congenital long QT syndrome is due to inherited mutations among genes that code voltage gated ion channels contributing to the cardiac action potential (101). Those affected exhibit a prolonged QT interval and are prone to arrhythmic death. Medications such as terfenadine and sotalol can cause acquired QT prolongation (102).

Lengthening of the QT interval is induced by both experimental and clinical hypoglycemic episodes (Figure 1-4), in nondiabetic individuals and

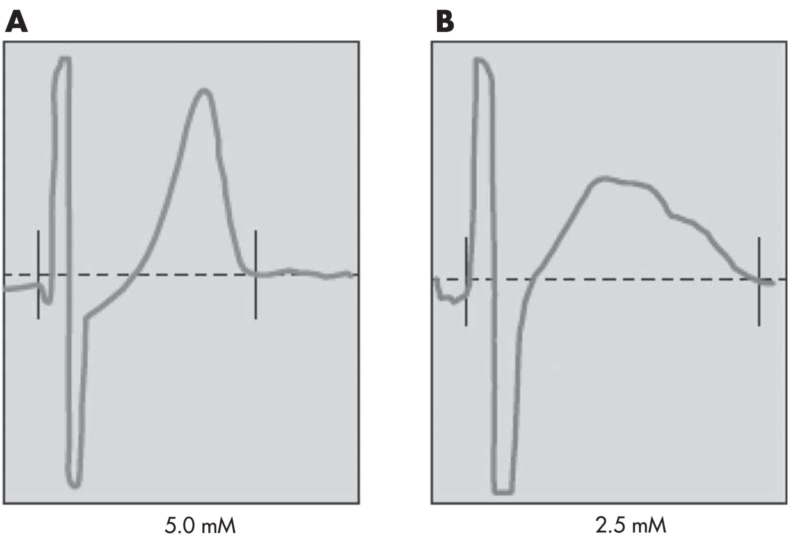

FIG 1-4. Typical electrocardiographic changes in response to hypoglycemia (right panel), compared to euglycemia (left panel) in a non-diabetic volunteer. (From Marques et al, 1997) (103).

patients with either type 1 or type 2 diabetes (103, 104), and in both adults (105) and children (106). Reductions in serum potassium and sympatho-adrenal activation both contribute to the effect (107). Thus one possible mechanism accounting for sudden death in type 1 diabetes is acquired long QT syndrome caused by insulin-induced hypoglycemia (108).

An additional factor that could contribute to arrhythmic risk during hypoglycemia is the contribution of autonomic activation, particularly if the parasympathetic and sympathetic components are affected differentially. Parasympathetic activation (or suppression of sympathetic activation) will cause bradycardia, an arrhythmia reported during nocturnal episodes (109). Studies using heart rate variability to measure cardiac sympathetic and parasympathetic tone have reported inconsistent changes during hypoglycemia (110–112), and the additional effect of autonomic neuropathy is also uncertain (113).

A recent case report of a sudden death during hypoglycemia in a young male patient with type 1 diabetes during coincidental continuous glucose monitoring is strong additional evidence of an arrhythmic death (114), but the precise mode of death remains unclear. It is likely that both genetic and environmental affects combine together, but important areas of uncertainty, including the factors that are most important and whether it is possible to identify those at greatest risk, are still to be determined.

Hypoglycemia as a Risk Factor for Increased All-Cause and Cardiovascular Mortality

Recent trials of intensive glycemic control in individuals with type 2 diabetes have also suggested that hypoglycemia might contribute to mortality. Severe hypoglycemic events in trial participants appear to be associated with increased risk of mortality in the months following the event compared to those with no history of severe episodes. The ACCORD (Action to Control Cardiovascular Risk in Diabetes), ADVANCE (Action in Diabetes and Vascular Disease: Preterax and Diamicron MR Controlled Evaluation) and VADT (Veterans Affairs Diabetes Trial) have in total randomized around 24,000 patients at increased cardiovascular risk to intensive glucose control (115–117). In 2008, the ACCORD trial was terminated early due to increased mortality in the intensive treatment arm, which equated to 54 excess deaths during the 3 years of the trial. Hypoglycemia (proportionately greater in those treated intensively) was among several potential causes, including weight gain, the effect of medication, or even chance. In ACCORD, participants who had experienced at least 1 severe event had higher mortality ratio than those with no hypoglycemia in both arms, with a hazard rate of 1.41 (118). Interestingly, there was also a relationship between a severe hypoglycemic episode and downstream mortality in both VADT and ADVANCE. However, although all 3 trials have reported a similar association; they could not establish a causal relationship. In ADVANCE, similar associations were also reported with other outcomes, including microvascular complications and noncardiac mortality (119). It seems equally possible that the risk of mortality might be due to residual confounding and be a marker of ill health and vulnerability.

Indeed the reported data linking hypoglycemia, mortality, and intensive glycemic control are inconsistent and apparently confusing. The ACCORD trial showed that the relationship between a severe hypoglycemic event and mortality was greater in those randomized to standard therapy (118). While this observation has been cited to refute a causal effect of hypoglycemia on mortality, the interpretation does not acknowledge the effect of tight control and repeated hypoglycemia, which is well known to suppress sympathoadrenal responses to subsequent hypoglycemic episodes. Thus if hypoglycemia has effects on cardiovascular risk through sympathoadrenal responses, then it might be expected that those experiencing occasional episodes in the standard group would demonstrate stronger sympathoadrenal activation with a more powerful effect on the cardiovascular system.

Evidence has also emerged recently of an adverse effect of intensive glucose control among nondiabetic individuals with critical illness. Original work in this area suggested that intensive glucose control reduced mortality in critically ill individuals admitted to a surgical intensive care unit, but subsequent trials have not been able to reproduce these findings. Indeed, a recent trial reported increased mortality in those treated intensively (120), and the markedly higher rates of hypoglycemia in this group have clear potential relevance to the observed outcomes (121). Subsequent post hoc analyses have confirmed the association between severe hypoglycemia and mortality, although a causal relationship cannot be established. Furthermore, the fact that the greatest rate of mortality was observed in those who were not taking insulin but who had experienced hypoglycemia suggests that confounding by vulnerability to hypoglycemia in those who are the most ill contributes to these findings (120, 122, 123).

In summary, despite a clear association between hypoglycemia and mortality, it is not yet possible to establish a causal relationship, and important confounders including increased vulnerability to both hypoglycemia and mortality probably contribute to these observations. Nevertheless, there are plausible mechanisms whereby hypoglycemia could increase cardiovascular risk. It seems appropriate to avoid very strict glucose targets or approaches more likely to cause hypoglycemia in individuals who are most vulnerable.

Potential Mechanisms by which Hypoglycemia Might Increase Vascular Risk

Although it has not been possible to prove a causal link between hypoglycemia and cardiovascular events, several experimental studies have demonstrated the potentially adverse effects of hypoglycemia on the cardiovascular system. Hypoglycemia leads to activation of the autonomic nervous system, an increase in circulating catecholamines, and hemodynamic changes. These will affect the heart and could conceivably worsen microvascular complications. For example, acute changes in peripheral blood flow could aggravate pre-existing renal disease, and sudden changes in intraocular pressure might precipitate vitreous hemorrhage in individuals with neovascularization. However, the relationship between hypoglycemia and pathophysiological changes that might worsen macrovascular disease is more firmly established (124). The pathway for the onset of acute coronary syndrome and myocardial infarction is well described, and it is striking how many components of the pathway can be affected by hypoglycemia.

Hypoglycemia and Macrovascular Complications

Direct precipitation of an acute cardiovascular event (either fatal or nonfatal myocardial infarction or stroke) by hypoglycemia would appear to be rare, yet experimental studies have demonstrated that hypoglycemia can stimulate mechanisms that both accelerate atherosclerosis and contribute to plaque instability. As described previously, hypoglycemia, by contributing to abnormal cardiac repolarization and disturbing autonomic function, might lower the threshold required to initiate a fatal arrhythmic event. Furthermore, sympathoadrenal activation will increase myocardial oxygen demand, which could also aggravate cardiac ischemia and provoke a cardiac event.

Experimental hypoglycemia can impair endothelial function through increases in von Willebrand factor (125) and endothelin levels (126, 127). Atherogenic cell adhesion molecules are expressed following hypoglycemia (127). However, the effects on the acute inflammatory response are less clear, with inconsistent responses following hypoglycemia on interleukin 6 and C-reactive protein (125, 127, 128).

Platelet activation and aggregability are increased following hypoglycemia (125, 127), a mechanism probably mediated by sympathoadrenal activation (129). Initiation of an acute coronary syndrome may also be related to the state of the fibrinolytic system, and this also appears to be activated by hypoglycemia (130). Other work has demonstrated increased coagulation and reduced fibrinolysis during sustained insulin-induced episodes (127)

The hemodynamic response following hypoglycemia leads to increases in heart rate, although this rarely rises above 100 beats per minute, perhaps because of simultaneous parasympathetic activation. Systolic blood pressure increases, accompanied by a fall in diastolic blood pressure, widening of pulse pressure, an increase in cardiac output, and a fall in peripheral vascular resistance. There is a transient increase in the elasticity of the arterial system (131), which is diminished in people with type 1 diabetes of longer duration (> 15 years), and premature arterial stiffening is common in diabetes of both types. Accelerating the return of the pressure wave generated during systole to the heart may diminish diastolic coronary filling. A fall in myocardial blood flow reserve (132) may be of limited clinical significance in a healthy individual but could be critical in a person with pre-existing coronary heart disease.

There are few reports of clinical hypoglycemic episodes initiating a cardiac ischemic event. Silent ischemia in a diabetic patient with suspected coronary heart disease has been reported (133), and one study utilizing continuous glucose monitoring reported ischemic electrocardiogram (ECG) changes during hypoglycemia (some being accompanied by chest pain) that were not observed when glucose was normal or high (134).

Socioeconomic Consequences

Driving

Because hypoglycemia affects cognitive function and mood, it can affect many everyday activities that require the interaction of several cognitive domains. One such important activity is driving; progressive neuroglycopenia interferes with driving performance and can cause road traffic accidents. To investigate the effect on driving performance, hypoglycemia was induced in adults with type 1 diabetes in a driving simulator, which had visual, auditory, and kinesthetic feedback to track several performance variables (135, 136). Driving performance began to deteriorate when blood glucose declined below 3.8 mmol/L (68 mg/dL), manifested by inappropriate speed and braking, poor road positioning, driving off the road, ignoring road signs and traffic lights, and crashes. Only 1 in 3 drivers self-treated their low blood glucose while driving, and then not until it had fallen to 2.8 mmol/L (50 mg/dL) or less. These seminal studies also demonstrated that driving utilized a substantial amount of infused dextrose and has a significant metabolic demand (137), so that a prophylactic snack should be eaten before driving when blood glucose is below 5.0 mmol/L (90 mg/dL). Hypoglycemia may impair visual perception during driving, particularly under conditions of limited perceptual time and low visual contrast (poor light), as it affects visual information processing and causes a deterioration in contrast sensitivity, inspection time, visual change detection, and visual movement detection (138, 139).

While hypoglycemia can adversely affect driving performance, all drivers with diabetes do not exhibit a similar degree of risk, and it may be difficult to identify which individual drivers are most vulnerable. Factors that have been shown to increase driving risk include previous episodes of severe hypoglycemia (140, 141), previous hypoglycemia while driving (140), strict glycemic control (low HbA1c) (141), and a failure to test blood glucose before driving (142). Surprisingly, impaired awareness of hypoglycemia has seldom emerged as a major predictor for driving-related accidents. Affected individuals may be initially resistant to cognitive dysfunction during moderate hypoglycemia (blood glucose 2.5 mmol/L (45 mg/dL)) and recover more quickly than those with intact awareness (63). Furthermore, accident risk can be minimized by frequent measurement of blood glucose. Nonetheless, blood glucose awareness training for people with impaired awareness of hypoglycemia has been reported to reduce their subsequent rate of motor vehicle accidents (143). The American Diabetes Association has published a position statement with recommendations about individual evaluation of risks to driving associated with diabetes (144).

Drivers with type 1 diabetes suffer more motor vehicle accidents than drivers with type 2 diabetes treated with insulin (142). However, hypoglycemia-related road traffic accidents are not exclusively associated with insulin therapy. Drivers with type 2 diabetes (aged <65 years) receiving treatment with medications other than insulin, who had a history of severe hypoglycemia, recorded a higher rate of motor vehicle accidents than those with no hypoglycemia (145). Ongoing research is essential to inform driving licensing authorities how to identify those drivers who are at greatest risk of having motor vehicle accidents and who should therefore be subject to licensing restrictions. At present, these vary between countries and continents, and influence employment prospects for people with insulin-treated diabetes.

Recommended safe driving practices include blood glucose testing before driving and at regular intervals on longer journeys, and keeping a supply of both fast-acting and complex carbohydrate in the vehicle. If hypoglycemia occurs while driving, the driver should stop the vehicle and should not recommence driving for 30 to 45 minutes after the low blood glucose has been treated, as restoration of normal cognitive function is delayed. Unfortunately, these simple measures are frequently ignored (146), and many health care professionals remain uninformed about the precautions required for safe driving (147). Routine education is essential for both groups (Box 1-1).

Employment

Some occupations are not available to people who require insulin therapy because of the potential risk of developing hypoglycemia. With some jobs, particularly those involving public transport or jeopardizing the safety of colleagues, any risk of hypoglycemia is considered unacceptable. In other situations, employers may impose restrictions on employment, often based on advice from occupational health physicians. In general, people with insulin-treated diabetes are not permitted to work in isolated or dangerous areas or at unprotected heights. They are usually barred from serving in the armed forces, because the provision of insulin and an appropriate diet may be impossible during periods of armed conflict. Employment is not usually available as emergency workers, civil aviation (with some exceptions), the offshore oil industry, and in many forms of commercial driving (148).

Box 1-1. Advice for drivers with insulin-treated diabetes regarding hypoglycemia

• If hypoglycemia occurs while driving, stop the vehicle in a suitable location; leave the driver's seat.
• Always keep an emergency supply of readily accessible fast-acting carbohydrate (e.g. glucose tablets or sweets) *in the vehicle.*
• Check blood glucose before driving (even on short journeys) and estimate at regular intervals on long journeys.
• Take regular meals and snacks, and rest periods on long journeys; avoid alcohol.
• If hypoglycemia is experienced, do not drive until 45 minutes after blood glucose is restored to normal (delayed recovery of cognitive function).
• Carry personal identification indicating 'diabetes' in case of injury in a motor vehicle accident.

Severe hypoglycemia does not appear to be a major problem in the workplace among most employees with type 1 diabetes (149). However, many people with insulin-treated type 2 diabetes have reported problems in relation to their work following mild hypoglycemia, including not going to work or arriving late because hypoglycemia (particularly nocturnal) had occurred outside working hours or it affected work performance to cause significant loss of productivity (150).

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